

THE ARITHMETIC DEFECT OF OFF-LINE ZEROS: PARITY, THE MODIFIED EXPLICIT FORMULA, AND TWO-SCALE STRUCTURE

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ABSTRACT. We study the exact arithmetic footprint that a *finite* set of off-critical-line zeros of the Riemann zeta function would leave. The off-line zeros of ζ form complete Klein four-orbits $\{\frac{1}{2} \pm b \pm i\gamma\}$, and the negative index κ of the localized Weil form is the dimension of its maximal negative subspace. We first prove the geometry of that subspace: each orbit contributes exactly two linearly independent negative directions, so $\kappa = 2m$ is even, with the minimal non-trivial value 2; directions from orbits at distinct heights are exponentially nearly orthogonal, so the negative subspace decomposes as an approximate orthogonal sum of Lorentz planes. Assuming a finite negative index (Hypothesis D, $0 < \kappa = 2m < \infty$, strictly weaker than the negation of the Riemann Hypothesis), the completed zeta factors as $\xi = \xi_0 P_{4m}$ with P_{4m} the degree- $4m$ polynomial of the off-line zeros and ξ_0 an entire function sharing the functional equation and having all zeros on the line. This yields an *exact modified explicit formula* $\psi(x) = \psi_0(x) - D_m(x)$, where the arithmetic defect D_m is a finite sum of $4m$ oscillations whose amplitudes, frequencies, and phases are explicit; its dominant term grows like $x^{1/2+b_{\max}}$ and its energy like $X^{2b_{\max}}$. We then locate the precise obstruction: the companion $\zeta_0 = \xi_0/(\pi^{-s/2}\Gamma(s/2))$ has poles inside the critical strip, so by Landau's theorem it admits no Dirichlet series with non-negative coefficients and no Euler product — the analytic factorization is incompatible with the multiplicative structure of ζ . This is the wall we name (RSRP): the Euler product is incompatible with a finite non-zero index; it follows from the Riemann Hypothesis, does not imply it, and is distinct from the positivity, de Branges, and pair-correlation walls. Finally we exhibit the two-scale signature of a single orbit — an exponentially growing oscillation in the Chebyshev variable and a localized logarithmic depression of $\log |\zeta(\frac{1}{2} + it)|$ near $t = \gamma$ — prove that both are compatible with the unconditional prime number theorem and (in the polynomially-bounded regime) with the multiplicative-chaos model, and state precisely which three inputs any proof of the exclusion principle $\kappa \in \{0, \infty\}$ would require, none of which is currently available. Nothing here assumes or proves the Riemann Hypothesis.

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1. INTRODUCTION

The Riemann Hypothesis (RH) is the assertion that every nontrivial zero $\rho = \beta + i\gamma$ of ζ has $\beta = \frac{1}{2}$. Suppose it fails. What, precisely, would a finite collection of off-line zeros do to the distribution of primes, and where exactly does the assumption that they exist collide with the arithmetic of ζ ? This paper answers both questions exactly, under the working hypothesis that the index is finite.

We write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function, $\mathcal{Z}_{\text{off}} = \{\rho : \beta_\rho \neq \frac{1}{2}\}$ for the off-line zeros and $\mathcal{Z}_{\text{cl}} = \{\rho : \beta_\rho = \frac{1}{2}\}$ for the on-line ones. The functional equation $\xi(s) = \xi(1-s)$ and the reality $\overline{\xi(\bar{s})} = \xi(s)$ make $\mathcal{Z}_{\text{off}}$ invariant under the Klein four-group $V = \{1, \sigma, \iota, j\}$, $\sigma(s) = \bar{s}$, $\iota(s) = 1-s$, $j = \sigma\iota$, so the off-line zeros come in quartets

$$\mathcal{O}_j = \{\sigma_j \pm i\gamma_j, (1 - \sigma_j) \pm i\gamma_j\}, \quad \sigma_j = \frac{1}{2} + b_j > \frac{1}{2}, \quad \gamma_j > 0.$$

The localized Weil form Q — the quadratic form whose positivity is RH — is indefinite, and its negative (Kreĭn) index κ is the dimension of its maximal negative subspace $\mathcal{N}$; one has $\text{RH} \iff \kappa = 0$. The companion functional-analytic study of Q [1] localizes κ in the off-axis evaluation functionals attached to off-line zeros and identifies κ with the number of zeros of the structure function $E_\xi(z) = \xi(\frac{1}{2} + iz)$ in one open half-plane. The present paper takes that index as given and develops its *arithmetic* content.

The standing hypothesis throughout the conditional parts is:

Hypothesis D. $0 < \kappa = 2m < \infty$: there are finitely many off-line zeros, forming $m \geq 1$ Klein quartets.

Hypothesis D is strictly weaker than $\neg\text{RH}$: it is the negation of RH *together with* finiteness of the off-line set. Finiteness is itself unconditionally open and weaker than RH (it is not delivered by the positive-proportion or zero-density theorems [1]); we assume it only to make the defect a finite, fully explicit object, and we are careful to mark every statement that uses it.

The paper has three movements. §2 proves the geometry of $\mathcal{N}$: parity $\kappa = 2m$, angular separation between orbits, and the Lorentz-plane decomposition. §3–§5 derive, under Hypothesis D, the factorization $\xi = \xi_0 P_{4m}$ and the exact modified explicit formula $\psi = \psi_0 - D_m$, analyze the defect D_m , and prove the arithmetic-analytic gap that the companion ζ_0 has no Euler product — the wall (RSRP). §6–§7 exhibit the two-scale signature of an orbit, prove its compatibility with the unconditional prime number theorem and with multiplicative chaos in the polynomially-bounded regime, and state the barrier theorem isolating the three inputs that a proof of the exclusion principle $\kappa \in \{0, \infty\}$ would require. None of this assumes or proves RH.

2. THE NEGATIVE SUBSPACE: PARITY AND LORENTZ GEOMETRY

We work in the reproducing-kernel Kreĭn space $\mathcal{H}$ of test functions on which Q acts, with negative subspace $\mathcal{N} = \{v : Q(v, v) < 0\}$, $\dim \mathcal{N} = \kappa$ [1]. For an off-line orbit $\mathcal{O}_j$ write $\rho_j^+ = \sigma_j + i\gamma_j$, $\rho_j^- = (1 - \sigma_j) + i\gamma_j$, and split the orbit by height into the upper sub-orbit $\mathcal{O}_j^+ = \{\rho_j^+, 1 - \bar{\rho}_j^+\}$ (both at height $+\gamma_j$) and the lower sub-orbit $\mathcal{O}_j^- = \{\bar{\rho}_j^+, 1 - \rho_j^+\}$ (both at height $-\gamma_j$).

Proposition 2.1 (The form restricted to one sub-orbit). *For test functions whose transform is concentrated near $+\gamma_j$, the contribution of $\mathcal{O}_j^+$ to the Weil form is*

$$Q|_{\mathcal{O}_j^+}(f, f) = 4[(\text{Re } \widehat{f}(\rho_j^+))^2 - (\text{Im } \widehat{f}(\rho_j^+))^2],$$

a two-dimensional form of signature $(1, 1)$ whose unique negative direction v_j^+ is characterized by $\text{Re } \widehat{f}(\rho_j^+) = 0$. The lower sub-orbit $\mathcal{O}_j^-$ gives the analogous form near $-\gamma_j$, with unique negative direction v_j^- .

Proof. The contribution of $\mathcal{O}_j^+ = \{\rho_j^+, 1 - \bar{\rho}_j^+\}$ to $\sum_\rho \widehat{f}(\rho)^2$ is $\widehat{f}(\rho_j^+)^2 + \widehat{f}(1 - \bar{\rho}_j^+)^2$. The functional-equation/reality symmetries give $\widehat{f}(1 - \bar{s}) = \overline{\widehat{f}(s)}$, so $\widehat{f}(1 - \bar{\rho}_j^+) = \overline{\widehat{f}(\rho_j^+)}$; writing $\widehat{f}(\rho_j^+) = u + iv$, the sum is $(u + iv)^2 + (u - iv)^2 = 2(u^2 - v^2)$, i.e. $4(u^2 - v^2)$ after the archimedean normalization. The lower sub-orbit follows by conjugation. $\square$

Proposition 2.2 (Sub-orbit and inter-orbit orthogonality). *Let $f^\pm$ have transforms supported in a δ -neighborhood of $\pm\gamma_j$ in the imaginary direction, $\delta < \gamma_j$. Then $Q(f^+, f^-) = O(e^{-c\gamma_j^2/\delta^2})$. More generally, negative directions localized at distinct heights $\gamma_j \neq \gamma_k$ satisfy $Q(v_j, v_k) = O(e^{-c(\gamma_j - \gamma_k)^2/\delta^2})$.*

Proof. $Q(f^+, f^-) = \sum_\rho \widehat{f}^+(\rho) \widehat{f}^-(\rho) - B_{\text{arith}}(f^+, f^-)$. The archimedean cross term involves $\int \widehat{f}^+ \widehat{f}^- \Omega$ with Ω smooth; the supports are disjoint, so it vanishes. In the zero sum, a zero at height $\sim +\gamma_j$ contributes $|\widehat{f}^-(\rho)| = O(e^{-c\gamma_j^2/\delta^2})$ by concentration, and symmetrically; summing over zeros (Riemann–von Mangoldt density) gives the bound. The inter-orbit estimate is identical with $\gamma_j - \gamma_k$ in place of γ_j . $\square$

Theorem 2.3 (Parity of the index). $\kappa \equiv 0 \pmod{2}$; equivalently $\kappa = 2m$, with m the number of off-line Klein quartets. If RH fails, then $\kappa \geq 2$: the possible values of κ are $\{0, 2, 4, \dots\}$, and no odd value is consistent with the symmetry of ζ . The same holds for any primitive real Dirichlet L -function. With multiplicities, a zero of order ℓ contributes 2ℓ to κ .

Proof. Each orbit $\mathcal{O}_j$ contributes, by Proposition 2.1, one negative direction v_j^+ from $\mathcal{O}_j^+$ and one v_j^- from $\mathcal{O}_j^-$. These are independent: $Q(v_j^+, v_j^-) = O(e^{-c\gamma_j^2/\delta^2})$ by Proposition 2.2 with $\delta < \gamma_j$, strictly smaller than $\sqrt{Q(v_j^+, v_j^+) Q(v_j^-, v_j^-)}$, so the two are not proportional. The archimedean term and prime sum are non-negative, so every negative direction arises from an off-line evaluation and is accounted for by this orbit structure; hence $\dim \mathcal{N} = 2m$. This agrees with the structure-function count: the negative index equals the number of zeros of E_ξ in one open half-plane, which is exactly the two members of each quartet on a fixed side, i.e. $2m$ [1]. $\square$

Remark 2.4 (Why two, and where the count is robust). The two negative directions of an orbit live at the two heights $\pm\gamma_j$ and are independent in the full (complex) Kreĭn space $\mathcal{H}$; on a space of even real test functions the reality $\widehat{f}(\bar{z}) = \overline{\widehat{f}(z)}$ identifies the two heights and a quartet manifests there as a single Lorentzian block. The robust, representation-independent count is the structure-function one, $\kappa = \#\{\text{zeros of } E_\xi \text{ in one half-plane}\} = 2m$ [1]; the height decomposition above is the geometric realization of those two zeros.

Theorem 2.5 (Lorentz-plane decomposition). *Under Hypothesis D the negative subspace decomposes as an approximate orthogonal direct sum of m Lorentz planes,*

$$\mathcal{N} \approx \bigoplus_{j=1}^m \mathbb{R}_j^{1,1},$$

where $\mathbb{R}_j^{1,1} = \text{span}\{w_j^+, v_j^+\}$ carries signature $(1, 1)$ (w_j^+ the positive, v_j^+ the negative direction of $\mathcal{O}_j^+$), and the planes at distinct heights are exponentially nearly orthogonal in Q (Proposition 2.2). The approximation is exact in the limit of well-separated heights.

Proof. Each orbit contributes the signature- $(1, 1)$ plane of Proposition 2.1 at height $+\gamma_j$ (and its mirror at $-\gamma_j$); by Proposition 2.2 the Gram matrix between planes at $\gamma_j \neq \gamma_k$ is $O(e^{-c(\gamma_j - \gamma_k)^2/\delta^2})$, so the total form is block-diagonal up to exponentially small couplings. $\square$

Corollary 2.6 (Parity-gap trichotomy). *The index satisfies $\kappa \in \{0\} \cup 2\mathbb{Z}_{>0} \cup \{\infty\}$. Combined with the exclusion of the finite non-zero (Pontryagin) regime by the finite-defect analysis [1], the operative dichotomy is between $\kappa = 0$ (RH) and the arithmetically-loaded regimes $2\mathbb{Z}_{>0} \cup \{\infty\}$ studied below. We do not assert $\kappa \neq$ any particular value; Theorem 2.3 only fixes the parity.*

3. THE MODIFIED EXPLICIT FORMULA

We now assume Hypothesis D and read off the arithmetic consequences. The finiteness of $\mathcal{Z}_{\text{off}}$ makes the off-line part of the zero sum a genuine finite object.

Proposition 3.1 (Factorization). *Under Hypothesis D,*

$$\xi(s) = \xi_0(s) P_{4m}(s), \quad P_{4m}(s) = \prod_{\rho \in \mathcal{Z}_{\text{off}}} (s - \rho),$$

where P_{4m} is a degree- $4m$ polynomial with conjugate-symmetric roots and ξ_0 is entire, shares the functional equation $\xi_0(s) = \xi_0(1-s)$, and has all its zeros on $\text{Re } s = \frac{1}{2}$.

Proof. Divide the Hadamard product of ξ by the finite product over $\mathcal{Z}_{\text{off}}$; the quotient ξ_0 is entire of order one with zero set $\mathcal{Z}_{\text{cl}}$. The set $\mathcal{Z}_{\text{off}}$ is V -invariant, so P_{4m} is invariant under $s \mapsto 1-s$ and under conjugation, whence $\xi_0 = \xi/P_{4m}$ inherits the functional equation and reality. $\square$

Definition 3.2 (Arithmetic defect and virtual Chebyshev function). Under Hypothesis D set

$$D_m(x) := \sum_{\rho \in \mathcal{Z}_{\text{off}}} \frac{x^\rho}{\rho}, \quad \psi_0(x) := x - \sum_{\rho \in \mathcal{Z}_{\text{cl}}} \frac{x^\rho}{\rho} - \log(2\pi) - \frac{1}{2} \log(1-x^{-2}).$$

D_m is real-valued ($\mathcal{Z}_{\text{off}}$ is conjugation-symmetric) and is a finite sum of $4m$ terms.

Theorem 3.3 (Modified explicit formula). *Under Hypothesis D, for every non-integer $x > 1$,*

$$\psi(x) = \psi_0(x) - D_m(x),$$

an exact identity. Writing $E(x) = \psi(x) - x$ and $E_0(x) = \psi_0(x) - x$, one has $E(x) = E_0(x) - D_m(x)$ with $E_0(x) = O(x^{1/2+\varepsilon})$ automatically (all zeros of ξ_0 are on the line); hence the dominant growth of E is carried entirely by the defect D_m .

Proof. Insert $\sum_\rho = \sum_{\mathcal{Z}_{\text{cl}}} + \sum_{\mathcal{Z}_{\text{off}}}$ into the classical explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \log(2\pi) - \frac{1}{2} \log(1-x^{-2})$. Under Hypothesis D the off-line sum is finite, needing no regularization; rearranging gives the identity. The bound on E_0 is the explicit formula for ξ_0 , whose zeros all lie on the line. $\square$

4. ANATOMY OF THE DEFECT

Fix an orbit $\mathcal{O}_j$ and write $\theta_j^\pm = \arg \rho_j^\pm$.

Proposition 4.1 (Single-orbit defect).

$$D_j(x) = \underbrace{\frac{2x^{1/2+b_j}}{|\rho_j^+|} \cos(\gamma_j \log x - \theta_j^+)}_{\text{dominant}} + \underbrace{\frac{2x^{1/2-b_j}}{|\rho_j^-|} \cos(\gamma_j \log x - \theta_j^-)}_{\text{subdominant}}.$$

Consequently $D_m(x) = \sum_{j=1}^m \frac{2x^{1/2+b_j}}{|\rho_j^+|} \cos(\gamma_j \log x - \theta_j^+) + O(x^{1/2-b_{\min}})$ as $x \rightarrow \infty$.

Proof. Pair conjugate zeros: $\frac{x^{\sigma_j+i\gamma_j}}{\sigma_j+i\gamma_j} + \frac{x^{\sigma_j-i\gamma_j}}{\sigma_j-i\gamma_j} = \frac{2x^{\sigma_j}}{|\rho_j^+|} \cos(\gamma_j \log x - \theta_j^+)$, and similarly for the $(1-\sigma_j)$ pair with exponent $1-\sigma_j = \frac{1}{2} - b_j$. Collecting the subdominant terms over all orbits gives the asymptotic. $\square$

Proposition 4.2 (Defect energy). *For the logarithmic energy $\mathcal{E}_j(X) = \int_X^{2X} |D_j(t)|^2 t^{-2} dt$,*

$$\mathcal{E}_j(X) \sim \frac{2^{2b_j} - 1}{|\rho_j^+|^2 b_j} X^{2b_j} \rightarrow \infty \quad (b_j > 0), \quad \sum_{j=1}^m \mathcal{E}_j(X) \sim C_m X^{2b_{\max}}.$$

Proof. The dominant part of $|D_j(t)|^2$ is $\frac{4t^{1+2b_j}}{|\rho_j^+|^2} \cos^2(\gamma_j \log t - \theta_j^+) + O(t)$; the cosine-square averages to $\frac{1}{2}$ (Riemann–Lebesgue), and $\int_X^{2X} t^{2b_j-1} dt = \frac{(2^{2b_j}-1)}{2b_j} X^{2b_j}$. The total is dominated by the largest b_j . $\square$

5. THE ARITHMETIC-ANALYTIC GAP AND THE WALL (RSRP)

The factorization $\xi = \xi_0 P_{4m}$ is purely *analytic*. We now show it is incompatible with the *multiplicative* structure of ζ . Set $\zeta_0(s) = \xi_0(s)/(\pi^{-s/2}\Gamma(s/2))$, the companion Dirichlet-type function; from $\xi'/\xi = \xi'_0/\xi_0 + P'_{4m}/P_{4m}$ one gets

$$-\frac{\zeta'_0(s)}{\zeta_0(s)} = -\frac{\zeta'(s)}{\zeta(s)} + \sum_{\rho \in \mathcal{Z}_{\text{off}}} \frac{1}{s - \rho}.$$

Lemma 5.1 (Arithmetic incapacity of ζ_0). *Under Hypothesis D:*

- (1) ζ_0 has a pole at each $\rho \in \mathcal{Z}_{\text{off}}$, hence poles inside the strip $0 < \text{Re } s < 1$;
- (2) ζ_0 is not representable as a Dirichlet series $\sum_n a_n n^{-s}$ with all $a_n \geq 0$;
- (3) ζ_0 admits no Euler product $\prod_p (1 - a_p p^{-s})^{-1}$ with $|a_p| \leq 1$;
- (4) the formal Λ_0 defined by $-\zeta'_0/\zeta_0 = \sum_n \Lambda_0(n) n^{-s}$ is not the von Mangoldt function of any standard arithmetic object.

Proof. (1) The poles of ζ_0 in the strip are the zeros of P_{4m} , i.e. $\mathcal{Z}_{\text{off}}$. (2) By Landau's theorem, a Dirichlet series with non-negative coefficients that converges somewhere has its abscissa of convergence as a real singularity and is otherwise holomorphic to its right — in particular it has no poles in $0 < \text{Re } s < 1$; (1) forbids this. (3) An Euler product of the stated form is holomorphic and zero-free for $\text{Re } s > 1$ and its continuation has no poles in $\text{Re } s > 0$; (1) again forbids it. (4) $\Lambda \geq 0$ produces non-negative Dirichlet coefficients, excluded by (2). $\square$

Remark 5.2. Lemma 5.1 is not a contradiction with Hypothesis D. It says the Kreĭn–Langer-type factorization is available in the analytic category (entire functions with a functional equation) but incompatible with the arithmetic category (multiplicative coefficients, Euler product) of ζ . The wall below is the assertion that this incompatibility is ultimately fatal to the existence of a finite non-zero index.

Definition 5.3 (The wall (RSRP)). *Resonant Spectral Rigidity of the Euler Product* (RSRP) is the statement: no entire function f satisfying the functional equation of ξ , carrying an Euler product $f(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \prod_p (1 - a_p p^{-s})^{-1}$ with $|a_p| \leq 1$, can have $0 < \kappa(f) < \infty$.

Theorem 5.4 ((RSRP) is a genuine, strictly weaker wall). (i) $\text{RH} \Rightarrow (\text{RSRP})$ (vacuously, $\kappa = 0$). (ii) $(\text{RSRP}) \not\Rightarrow \text{RH}$ in isolation: it excludes only the finite non-zero index, leaving $\kappa = \infty$ untouched, so it is consistent with infinitely many off-line zeros. (iii) (RSRP) is not a consequence of the positivity wall (an analytic condition on Q), the de Branges/global geometry wall (existence of a reproducing-kernel space), or the pair-correlation wall (a statistical hypothesis on spacings): it is an arithmetic incompatibility on the Euler product, of a different type from all three.

Proof. (i) is immediate. (ii): (RSRP) forbids $0 < \kappa < \infty$ but says nothing about $\kappa = \infty$, which is consistent with $\neg \text{RH}$; hence it cannot imply RH. (iii): the three established walls are, respectively, positivity of a quadratic form, existence of a geometric realization, and a spacing statistic; (RSRP) is the non-existence of a meromorphic-with-Euler-product object of finite index, an incompatibility between a polar divisor in the strip and multiplicativity. These are logically independent formulations. $\square$

6. TWO-SCALE STRUCTURE OF A SINGLE ORBIT

A single off-line orbit manifests at two distinct scales: an exponentially growing oscillation in the Chebyshev variable x (equivalently $t = \log x$), and a localized depression of $\log |\zeta(\frac{1}{2} + it)|$ near the ordinate $t = \gamma_j$.

Theorem 6.1 (Hadamard depression). *Under Hypothesis D, the contribution of $\mathcal{O}_j$ to $\log |\zeta(\frac{1}{2} + it)|$ is, up to $O(1)$ archimedean constants,*

$$\delta_j \log |\zeta(\frac{1}{2} + it)| = \log(b_j^2 + (t - \gamma_j)^2) + \log(b_j^2 + (t + \gamma_j)^2) - \log |\rho_j^+|^2 - \log |\rho_j^-|^2 + O(1).$$

In particular: at $t = \gamma_j$ the depth is $2 \log b_j + O(1)$; the depression is localized with half-width b_j ; and for $|t - \gamma_j| \gg b_j$ it is $O(\log(|t - \gamma_j|/b_j))$.

Proof. The four Hadamard factors $1 - s/\rho$ of $\mathcal{O}_j$ at $s = \frac{1}{2} + it$ have moduli $\sqrt{b_j^2 + (t \mp \gamma_j)^2}/|\rho_j^\pm|$; taking logs of their product gives the displayed formula (the convergence factors $e^{s/\rho}$ contribute $O(1)$). Setting $t = \gamma_j$ gives depth $2 \log b_j + O(1)$. $\square$

Proposition 6.2 (Average invisibility). *If $\gamma_j \in [T, 2T]$ then*

$$\frac{1}{T} \int_T^{2T} |\delta_j \log |\zeta(\frac{1}{2} + it)||^2 dt = O(b_j (\log b_j)^2 / T) \rightarrow 0.$$

The depression is invisible to the Cesàro average.

Proof. The integrand is $O((\log b_j)^2)$ on the window $[\gamma_j - b_j, \gamma_j + b_j]$ of width $2b_j$, and $O((\log |t - \gamma_j|)^2)$ outside with convergent tail; dividing by T gives the estimate. $\square$

Theorem 6.3 (Structure-function bounds and their compatibility). *Let $X(t) = (\psi(e^t) - e^t)e^{-t/2}$ and $S_2(\tau, T) = \frac{1}{T} \int_0^T |X(t + \tau) - X(t)|^2 dt$.*

(1) (Lower bound, under Hypothesis D.) *For $\tau \notin \frac{2\pi}{\gamma_j} \mathbb{Z}$,*

$$S_2(\tau, T) \geq \frac{4A_j^2 \sin^2(\gamma_j \tau / 2)}{2b_j} \cdot \frac{e^{2b_j T} - 1}{T} + O(e^{b_j T} / \sqrt{T}), \quad A_j = \frac{2}{|\rho_j^+|},$$

so $S_2(\tau, T) \gg e^{2b_j T} / T$.

(2) (Upper bound, unconditional.) *By Korobov–Vinogradov, $S_2(\tau, T) \leq C e^T e^{-2c_1 \sqrt{T}}$.*

(3) (Compatibility.) *The two bounds never contradict: $e^{2b_j T} / T \leq C e^T e^{-c_1 \sqrt{T}}$ holds for large T because $2b_j - 1 < 0$. A contradiction would require $b_j > \frac{1}{2}$, impossible since $\sigma_j < 1$.*

Proof. (1) Under Hypothesis D, $X = X_0 - Y$ with $Y(t) = D_m(e^t)e^{-t/2}$ and, by the on-line bound $\sum_{z_{cl}} |\rho|^{-2} < \infty$, the X_0 structure function is $O(1)$; the dominant term of $|Y(t + \tau) - Y(t)|^2$ from orbit j is $4A_j^2 e^{2b_j t} \sin^2(\gamma_j \tau / 2) \cos^2(\dots)$, and averaging $\cos^2 \rightarrow \frac{1}{2}$ with $\int_0^T e^{2b_j t} dt \sim e^{2b_j T} / (2b_j)$ gives the bound. (2) From $|\psi(x) - x| \leq Cx \exp(-c_1 \sqrt{\log x})$, $|X(t)| \leq C e^{t/2} \exp(-c_1 \sqrt{t})$, and square-integrating. (3) is the displayed inequality. $\square$

7. CHAOS COMPATIBILITY AND THE BARRIER THEOREM

The t -scale depression must coexist with the value distribution of ζ on the line. The Fyodorov–Hiary–Keating picture (established in part by Arguin–Bovier–Kistler, Najnudel, Radziwiłł–Soundararajan) gives, for t uniform in $[T, 2T]$, $\min_{[T, 2T]} \log |\zeta(\frac{1}{2} + it)| \sim -\log \log T + O(\log \log \log T)$.

Theorem 7.1 (Chaos compatibility is regime-dependent). *With $T \asymp \gamma_j$ the depression depth $2|\log b_j| + O(1)$ compares with the chaos minimum $\sim -c \log \gamma_j$ as follows:*

(1) *if $b_j \geq \gamma_j^{-C}$ (polynomially bounded below), the depth is $O(\log \gamma_j)$ and is accommodated — compatible;*

- (2) if $b_j = e^{-\gamma_j^\alpha}$, $\alpha \in (0, 1)$, the depth is $\sim 2\gamma_j^\alpha \gg \log \gamma_j$ — potentially incompatible;
- (3) if $b_j \leq e^{-c\gamma_j}$, the depth is $\sim 2c\gamma_j$ — incompatible with the chaos model.

Since no unconditional lower bound on b_j is known, uniform compatibility across all $b_j \in (0, \frac{1}{2})$ cannot be asserted.

Proof. The depth is Theorem 6.1; the minimum is the cited chaos result. In (1), $O(C \log \gamma_j)$ matches $O(-\log \gamma_j)$; in (2)–(3), γ_j^α and γ_j exceed $\log \gamma_j$ for large γ_j . $\square$

Remark 7.2 (A correction). Korobov–Vinogradov bounds b_j from *above* only ($b_j \leq \frac{1}{2} - c_0/(\log \gamma_j)^{2/3}$); it gives no lower bound, hence no upper bound on $|\log b_j|$. Any claim that the depression depth is $O(\log \log \gamma_j)$ is therefore unfounded: the correct conclusion is the regime-dependent Theorem 7.1.

Theorem 7.3 (Barrier theorem). *Any proof of the exclusion principle $\kappa \in \{0\} \cup \{\infty\}$ from the present results requires at least one of:*

- (F1) *an unconditional bound $\psi(x) - x = O(x^{1/2+\varepsilon_0})$ for some fixed $\varepsilon_0 < \min_j b_j$, independent of Hypothesis D;*
- (F2) *a property of the Euler product (valid for $\operatorname{Re} s > 1$) incompatible with poles of $-\zeta'/\zeta$ in $\operatorname{Re} s \in (\frac{1}{2}, 1)$ — i.e. the wall (RSRP);*
- (F3) *a property of the multiplicative-chaos model incompatible with a depression of $\log |\zeta(\frac{1}{2} + it)|$ of the depth permitted by Theorem 7.1.*

None of (F1), (F2), (F3) is currently available.

Proof. (F1) is necessary: without a better-than-Korobov–Vinogradov bound, the structure-function estimates are compatible (Theorem 6.3), so no contradiction arises from D_m . (F2) is necessary: the Euler product for $\operatorname{Re} s > 1$ gives no information on zeros in $\operatorname{Re} s \in (\frac{1}{2}, 1)$, and the standard candidate invariants (Selberg second moment, Mertens function, the Weil functional) are each either circular or reduce to a known wall. (F3) is necessary by Theorem 7.1. Their status: (F1) is equivalent to controlling zero locations, hence circular; (F2) is the wall (RSRP), unproven; (F3) is regime-dependent and cannot be closed without a lower bound on b_j . $\square$

8. CONCLUSION

Under the sole hypothesis that the off-line set is finite, a single Klein quartet of off-line zeros is a fully explicit object with a fully explicit footprint: two negative directions in the Weil form (Theorem 2.3), a Lorentz-plane block in the negative subspace (Theorem 2.5), an exact defect term D_m in the explicit formula (Theorem 3.3) growing like $x^{1/2+b_{\max}}$ with energy $X^{2b_{\max}}$ (Proposition 4.2), and a logarithmic depression of $\log |\zeta(\frac{1}{2} + it)|$ at $t = \gamma$ of depth $2|\log b|$ (Theorem 6.1). All of this is compatible with the unconditional prime number theorem (Theorem 6.3) and, in the polynomially-bounded regime, with the value distribution of ζ on the line (Theorem 7.1). The obstruction to excluding such a configuration is sharp and named: the analytic factorization $\xi = \xi_0 P_{4m}$ produces a companion with poles in the strip and hence no Euler product (Lemma 5.1), the incompatibility crystallized as the wall (RSRP) (Theorem 5.4); and any unconditional exclusion must supply one of three precisely identified inputs (Theorem 7.3), none of which is available. The contribution is a map of exactly where the finiteness of the index meets, and resists, the arithmetic of ζ . Nothing here assumes or proves the Riemann Hypothesis.

REFERENCES

- [1] D. A. Trejo Pizzo, *The localized Weil form and its Kreĭn–Pontryagin index: localization, obstruction, and rigidity*, preprint, 2026.
- [2] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice–Hall, Englewood Cliffs NJ, 1968.
- [3] H. Davenport, *Multiplicative Number Theory*, 3rd ed., GTM **74**, Springer, 2000.

- [4] E. Landau, *Über einen Satz von Tschebyschef*, Math. Ann. **61** (1905), 527–550.
- [5] A. Ivić, *The Riemann Zeta-Function: Theory and Applications*, Dover, 2003.
- [6] Y. V. Fyodorov, G. A. Hiary, J. P. Keating, *Freezing transition, characteristic polynomials of random matrices, and the Riemann zeta function*, Phys. Rev. Lett. **108** (2012), 170601.
- [7] L.-P. Arguin, D. Belius, A. J. Harper, *Maxima of a randomized Riemann zeta function, and branching random walks*, Ann. Appl. Probab. **27** (2017), 178–215.